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Key Points:
- The Qaidam basin experienced no obvious basin-scale rotation since Eocene
- The uniform paleomagnetic results support a relatively rigid Qaidam block
- The south side of the Kunlun Fault is the active side

Supporting Information:
- ReadMe
- Tables S1–S3

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Paleomagnetism of Eocene and Miocene sediments from the Qaidam basin: Implication for no integral rotation since the Eocene and a rigid Qaidam block

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Abstract The Qaidam basin is the largest topographic depression inside the Tibetan Plateau and it is a key factor to understanding the Cenozoic evolution of the northern Tibetan Plateau. Paleomagnetic data was obtained from the middle to late Eocene Xiaganchaigou Formation and the early to middle Miocene Xiayoushashan Formation from seven localities. The paleomagnetic results indicate that the Qaidam basin has not undergone obvious basin-scale vertical axis rotation with respect to the Eurasia Plate since the Eocene. Local clockwise rotation took place only at a few special locations along the northern margin of the Qaidam basin. The uniform paleomagnetic results at different localities support that the Qaidam basin is a relatively rigid block. Regional paleomagnetic and geodetic observations also suggest that crust south of the Kunlun fault moves eastward faster than crust north of the Kunlun fault.

1. Introduction

The Qaidam basin is the largest Cenozoic basin in the Tibetan Plateau. Located within the northern Tibetan plateau region, the evolution of the Qaidam basin has attracted researchers' widespread attention. The Qaidam basin is bounded by three huge fault zones: the Eastern Kunlun Fault Zone to the south, the Altyn Tagh Fault Zone to the northwest and the Qilian Shan-Nan Shan Thrust Zone to the northeast, and the relationship between these faults and the Qaidam basin is worth to study carefully. Compared to complexly deformed surrounding areas, the Qaidam basin shows only limited deformation of the Cretaceous to Tertiary sedimentary cover [*Meyer et al.*, 1998]. The change in elevation and intensity of deformation is generally attributed to the higher strength of the Qaidam crust [*Zhu et al.*, 1995; *Cook and Royden*, 2008; *Dayem et al.*, 2009], compared to surrounding regions that underwent intense compressional deformation and attendant crustal thickening.

Many researchers have attained a series of meaningful results within the Qaidam basin to reveal its geologic history [*Wang and Coward*, 1993; *Metivier et al.*, 1998; *Robinson et al.*, 2003; *Wang et al.*, 2004; *Zhu et al.*, 2006; *Yin et al.*, 2007, 2008a, 2008b; *Kent-Corson et al.*, 2009; *Kapp et al.*, 2011; *Heermance et al.*, 2013; *Song et al.*, 2013]. Moreover, an array of magnetostratigraphic and environmental magnetic work has also been conducted [*Sun et al.*, 2005; *Fang et al.*, 2007; *Lu et al.*, 2009; *Zhang et al.*, 2012a, 2012b; *Chang et al.*, 2012]. Specifically, some tectonic paleomagnetic results within the Qaidam basin have been published [*Chen et al.*, 2002; *Dupont-Nivet et al.*, 2002]. Despite these studies, there is still some disagreement concerning the timing, magnitude, and distribution of vertical axis rotations within the Qaidam basin. The mean paleomagnetic declinations at locations in central Qaidam basin indicate no vertical axis rotation since the Oligocene [*Dupont-Nivet et al.*, 2002]. In contrast, based on results from two other localities within the northern Qaidam basin, *Chen et al.* [2002] concluded that sinistral lateral-slip on the Altyn Tagh Fault was accommodated by approximately 20° of clockwise rotation of the Qaidam basin at the end of or after the Oligocene. To resolve this debate effectively, we conduct a paleomagnetic study of coeval Cenozoic rocks widespread in the Qaidam basin: the middle to upper Eocene Xiaganchaigou Formation and the lower to middle Miocene Xiayoushashan Formation. We sampled these units at seven localities throughout the basin (Xichagou, Gansen, Eboliang, Heishiqiu, Lulehe, Kushuiquan, and Gahai), which should provide an ideal opportunity to characterize the distribution of vertical axis rotations throughout the Qaidam basin. These sections cover most outcrops of these two formations inside the Qaidam basin (Figure 1).

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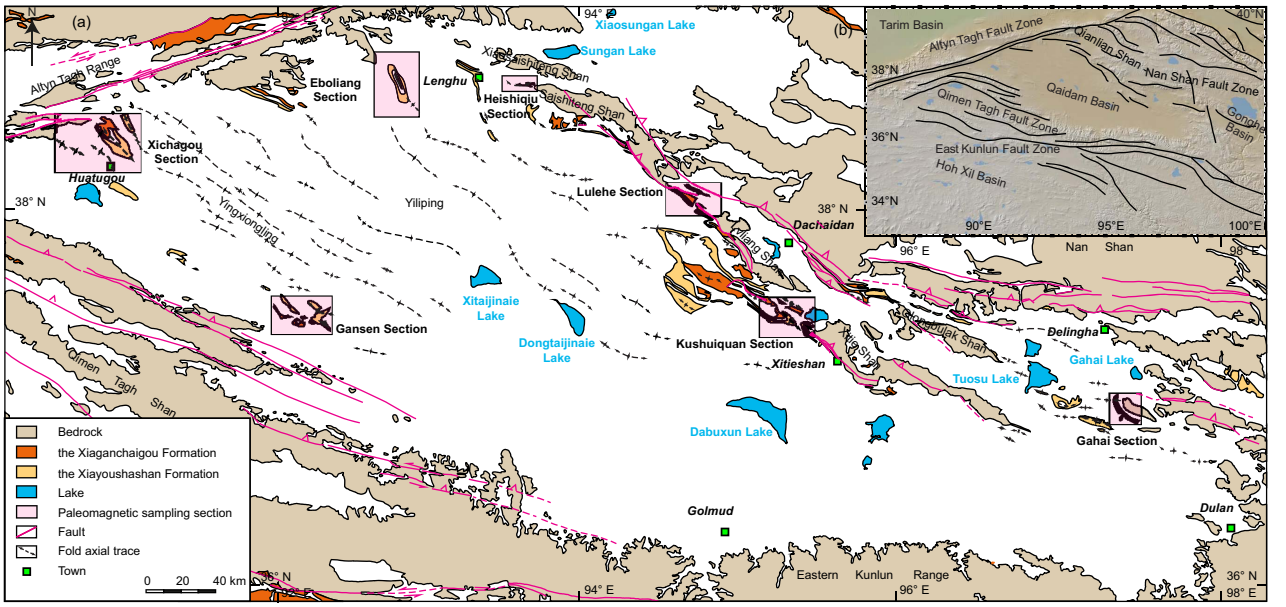


Figure 1. The geological map of the Qaidam basin and the sampling sections of the Xiaganchaigou and Xiayoushashan formations. (The small figure for tectonic background is based on *Yin et al*. [2007]).

2. Geological Background and Sample Collection

Most parts of the Qaidam basin are covered by Cenozoic sediments and preCenozoic strata outcrop only a little inside the basin, limited to the western Qaidam basin and the northern Qaidam basin. Major Cenozoic depositions in the Qaidam basin include: the Paleocene to early Eocene Lulehe Formation (E_{1+2}, 65∼49 Ma) [*Yang et al*., 1988; *Huo et al*., 1990; *Yang et al*., 1992; *Rieser et al*., 2006a, 2006b], the middle to late Eocene Xiaganchaigou Formation (E_3; 49∼32 Ma) [*Sun et al*., 1999, 2005], the Oligocene Shangganchaigou Formation (N_1, 32∼24 Ma) [*Sun et al*., 1999, 2005; *Lu et al*., 2009], the early to middle Miocene Xiayoushashan Formation (N_2^1, 24∼13 Ma) [*Sun et al*., 1999; *Fang et al*., 2007; *Lu et al*., 2009], the late Miocene Shangyoushashan Formation (N_2^2, 13∼7 Ma) [*Sun et al*., 1999; *Fang et al*., 2007], the Pliocene Shizigou Formation (N_2^3, 7∼2 Ma) [*Sun et al*., 1999; *Fang et al*., 2007], and the Pleistocene Qigequan Formation (<2 Ma) [*Sun et al*., 1999; *Fang et al*., 2007]. The evolution of the Qaidam basin is in close relationship with three basin-bounding fault zones: the Altyn Tagh Fault Zone (ATF), the Kunlun Fault Zone (KLF), and the Nan Shan Fault Zone (NSF). The ATF and KLF are famous strike-slip fault zones, while the NSF is characterized by a series of southwest-vergent thrusts. *Yin et al*. [2002, 2007, 2008a, 2008b] proposed that the northern basin-bounding structures, the NSF and the ATF, started coevally with the initial basin sedimentation before 49 Ma, while the southern basin-bounding structures, the KLF, started since the Oligocene. However, *Yue et al*. [2001] and *Zhuang et al*. [2011] proposed that the ATF initialed since the late Oligocene-earliest Miocene and the early Oligocene, respectively. Some researchers held the idea that the left-lateral strike-slip movement of the AFT obviously influenced the evolution of the NSF [*Wu et al*., 2012].

Seven paleomagnetic sampling localities were selected within the Qaidam basin for the middle to upper Eocene Xiaganchaigou Formation and the lower to middle Miocene Xiayoushashan Formation (Xichagou, XE, XN; Gansen, GE, GN; Eboliang, EE, EN; Heishiqiu, LE, LN; Lulehe, YE, YN; Kushuiquan, KE, KN; Gahai, BE, BN). Each sample group consists of eight sites, except the Xiayoushashan Formation at Gahai (BN, five sites) and the Xiayoushashan and Xiaganchaigou formations at Heishiqiu (LN, two sites and LE, four sites). At each sampling site, no <8 independently drilled samples are collected over several meters of stratigraphic thickness using a portable drill. All samples were oriented using a magnetic compass, and the location of each site was determined by a handheld GPS receiver.

3. Temperature-Dependent Susceptibility Testing and Results

Measurement of temperature-dependent susceptibility was measured using a KLY-4S Kappa bridge and CS-3 high-temperature furnace at the Paleomagnetism Geochronology Laboratory (PGL) of the Institute of

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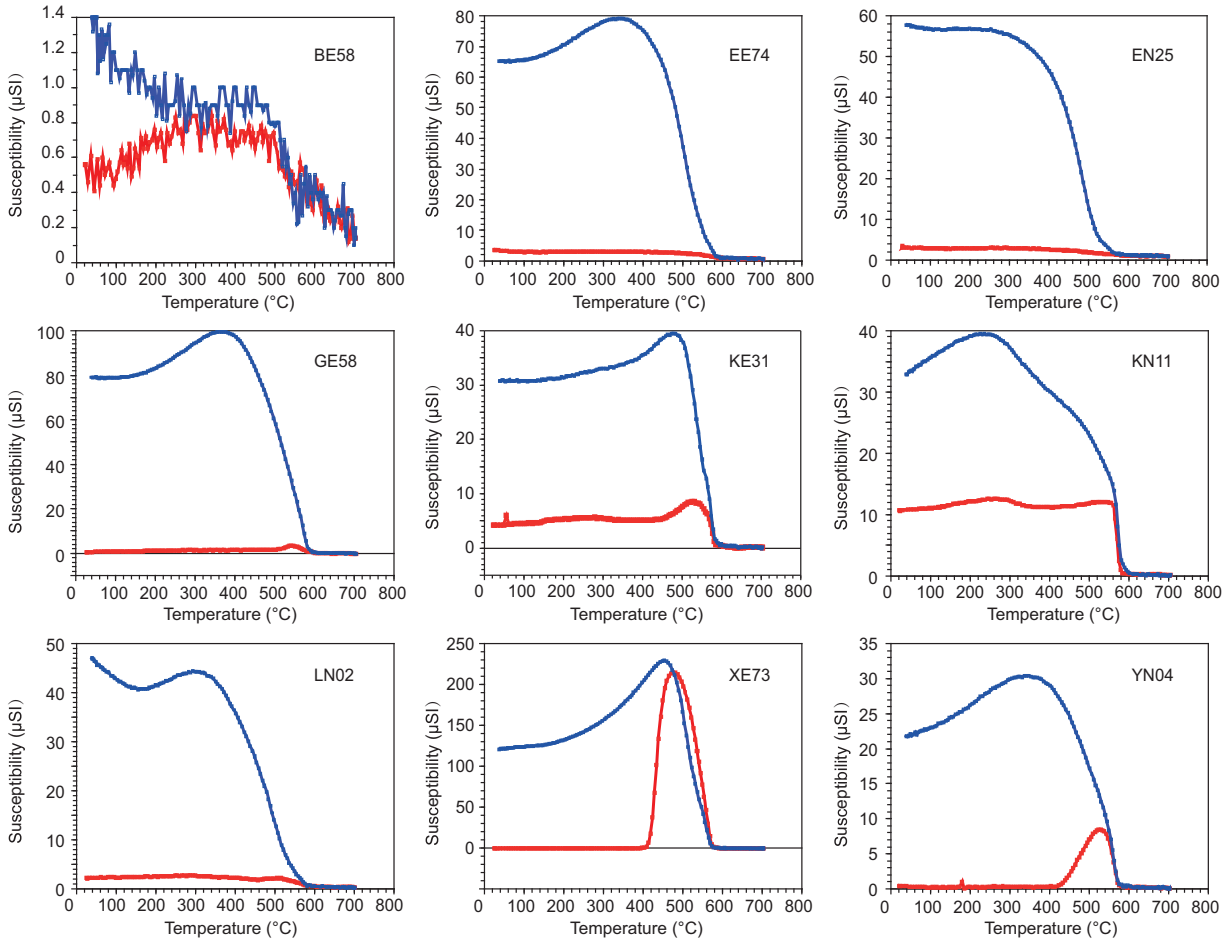


Figure 2. Temperature dependence of magnetic susceptibility (*k*-T curves) for representative samples. All curves were determined by heating from room temperature to ~700°C and cooling to room temperature in an argon atmosphere (Red line: heating curve and Blue line: cooling curve).

Geology and Geophysics, Chinese Academy of Sciences (CAS), obtained by continuous exposure of samples through temperature cycles from room temperature to 700°C and back to room temperature in an argon atmosphere. The mass of each specimen is around 300 mg. Thirty-four specimens were measured and nine representative results are presented here (Figure 2). The original curves were subtracted by measured furnace correction. Most samples show irreversible behaviors during thermomagnetic analysis. Except for specimen XE73, all the cooling cycles are greater than the heating cycles, revealing the forming of the new magnetic minerals with high susceptibility. The high-susceptibility magnetic minerals may come from the decomposition of iron-bearing silicate and argillaceous minerals or the dehydration of iron hydroxides [*Roberts et al.*, 1995; *van Velzen and Dekkers*, 1999; *Pan et al.*, 2000; *Deng et al.*, 2004]. Most of the temperature-dependent susceptibility curves are characterized by a dramatic decrease at about 580°C, revealing the existence of magnetite. Many samples also see a gradual decrease around 680°C, indicating hematite in some of the sample groups. Given that the susceptibility of magnetite is approximately 1000 times greater than hematite, we conclude that hematite is locally the main magnetic carrier for some specimens, such as BE58. Some heating curves decrease dramatically with initial heating, consistent with the contribution of a large amount of goethite, such as in specimens EE74 and EN25. On the other hand, the heating curves of some samples, such as KN11 and LN02, increase from room temperature to 300°C and decrease from 300 to 400°C, revealing a peak at around 300°C, probably revealing the existent of iron sulphide or a cracked maghemite rind around a magnetite core. Another obvious peak occurs around 500°C in the heating curves of specimens GE58, KE31, XE73, and YN04, which may be explained by new magnetic minerals with high susceptibility. We also note that heating specimen XE73 produced a foul sulfur odor, providing additional evidence for the presence of iron sulphide.

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4. Paleomagnetic Method and Results

The drilled samples were trimmed to 2.2 cm long specimen and the natural remanent magnetization (NRM) was measured by 2G Enterprises 755R cryogenic magnetometer at the PGL. Specimens were thermally demagnetized in intervals from 50 to 100°C at the lower temperatures to 10–20°C at higher temperatures. All measurements were completed in a magnetically shielded room with a maximum internal magnetic field <300 nT. Demagnetization results were evaluated using orthogonal diagrams [*Zijderveld*, 1967] and stereographic projections and magnetic components were analyzed by principal component analysis [*Kirschvink*, 1980]. The means of the observations were calculated using Fisher statistics [*Fisher*, 1953] and the great-circle approach of *McFadden and McElhinny* [1988]. Samples yielding maximum angular deviation (MAD) >15° were rejected for further analysis and site-mean directions with α_{95} > 25° were also rejected.

4.1. Xichagou

The Xichagou section is located at the northern tip of the Yingxiongling anticline in the western Qaidam basin, adjacent to the ATF. Samples from the Xiaganchaigou Formation (XE) were collected from the core and limbs of the anticline. The sediments are light cyan and yellow sandstone with straw yellow coarse sandstone. The Xiayoushashan Formation (XN) samples collected at both limbs of the anticline are light yellow, brownish yellow and orange mudstone, siltstone, and sandstone. The collected samples also include coarse sandstone with orange, green, and light grey siltstone and sandstone.

For the Xiaganchaigou Formation at Xichagou (XE), apart from a low unblocking temperature component (LTC), some thermal demagnetizations in orthogonal plots allow resolution of one high temperature component (HTC) with the unblocking temperatures mainly <600°C (Figure 3f). Most specimens are with high noise in the demagnetization path, and we only get a few good results. Therefore, the Xiaganchaigou Formation is combined with the Xiayoushashan Formation for future discussion. For the specimens from the Xiayoushashan Formation (XN), apart from a low temperature component (LTC), most thermal demagnetizations in orthogonal plots allow resolution of one high temperature component (HTC) mainly between 150–500°C and 500–680°C (Figure 3m).

The HTC components defined by the Xiaganchaigou and Xiayoushashan formations (XE and XN) yielded a nine site-mean direction of D = 347.4°, I = 14.3° before and of D = 352.5°, I = 47.2° after tilt correction (Figures 4o and 4p). The grouping of the data set is improved after tilt correction with the ratio k_s/k_g = 5.30 ($F(16,16)_{0.05}$ = 2.33), indicating a positive fold test [*McElhinny*, 1964], and the parametric progressive fold test also suggests a prefold origin (Figure 5d) [*Tauxe*, 2010].

4.2. Gansen

Gansen is located at the southernmost end of the Yingxiongling structure. The Xiaganchaigou Formation (GE) samples are brown, brownish red, light yellow, and brownish yellow mudstone with brown siltstone, whereas the Xiayoushashan Formation (GN) samples are mainly yellow and light yellow mudstone and siltstone.

For the specimens from the Xiaganchaigou Formation (GE) and the Xiayoushashan Formation (GN), apart from a low unblocking temperature component (LTC), most thermal demagnetization in orthogonal plots allow resolution of one HTC (Figures 3d and 3h). The seven site-mean observations of the Xiaganchaigou Formation (GE) yield a direction of D = 358.8°, I = 40.1° before and of D = 360.0°, I = 34.6° after tilt correction with slight improvement of data grouping following tilt correction yielding the ratio k_s/k_g = 1.39 (Figures 4g and 4h). The eight sites from the Xiayoushashan Formation (GN) provide a site-mean direction of D = 351.4°, I = 58.2° before and of D = 5.5°, I = 42.1° after tilt-correction with slight improvement of data grouping following tilt correction yielding the ratio k_s/k_g = 1.37 (Figures 4i and 4j).

The parametric progressive fold test for both the Xiaganchaigou Formation and the Xiayoushashan Formation (GE & GN) is positive, as the 95% confidence bounds on the degree of untilting required for maximizing concentration of the data τ_1 of orientation matrix are 62% and 124% (Figure 5b) [*Tauxe*, 2010]. The confidence bounds from the two data sets overlap in all three components, and the means of the reverse and normal models cannot be distinguished at 95% level of confidence, indicating a positive parametric reversal test for GE and GN (Figure 6b) [*Tauxe*, 2010].

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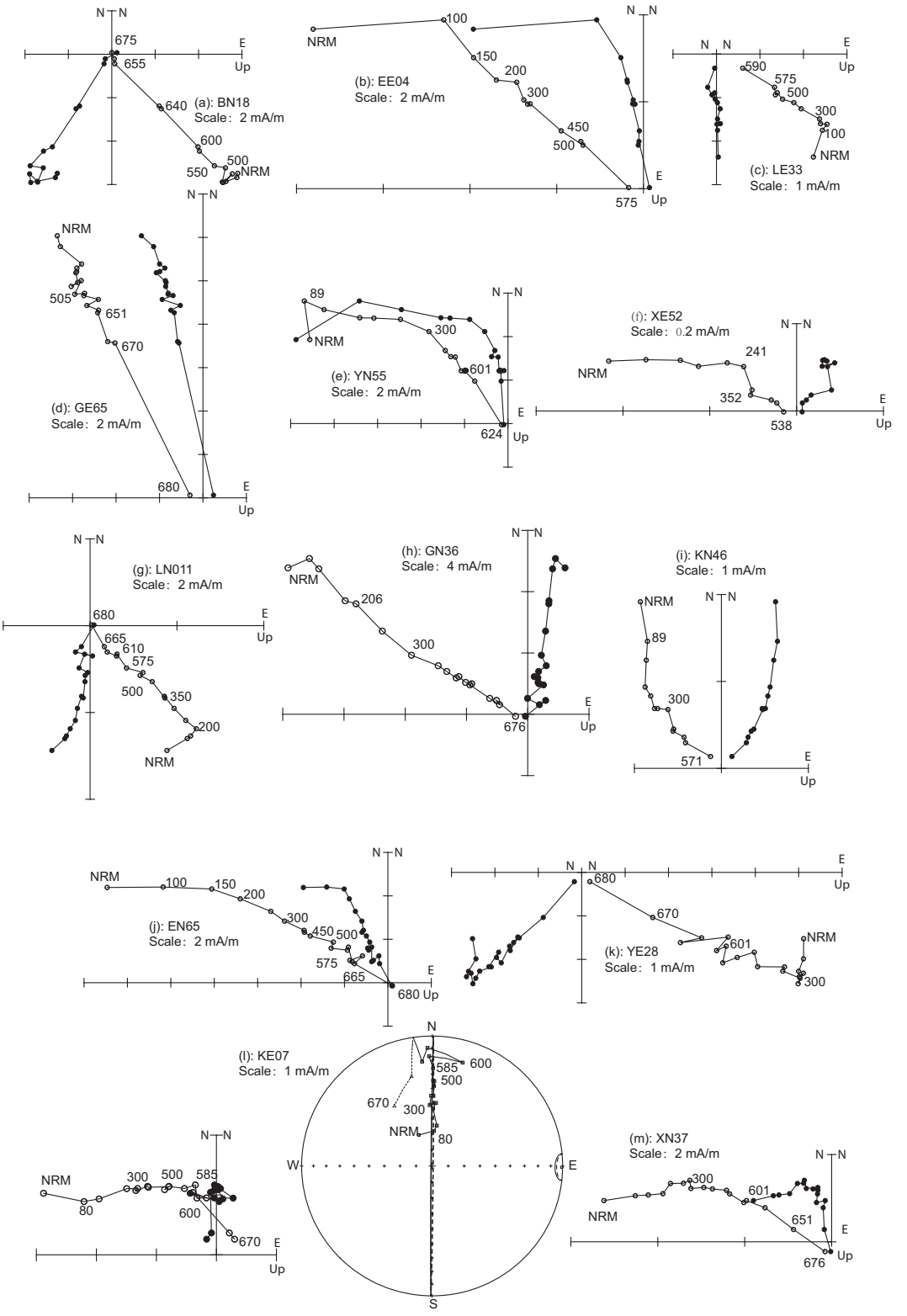


Figure 3. Orthogonal vector plots of representative specimens from each sample group. Directions are plotted in tilt-corrected coordinates. Solids and open circles present the vector endpoints projected onto horizontal and vertical planes.

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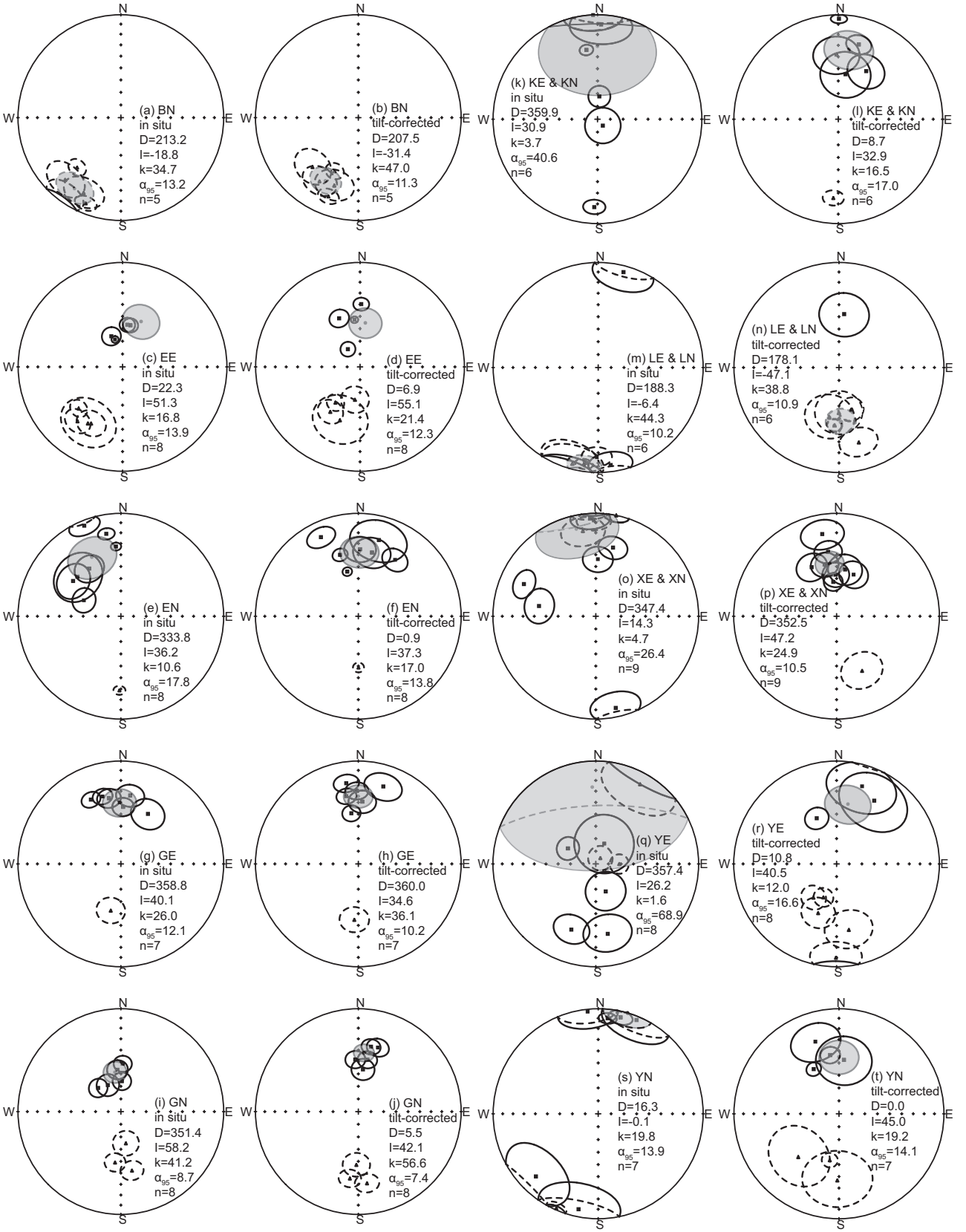


Figure 4. Equal-area projection of the site-mean high-temperature component (HTC) results (a, c, e, g, i, k, m, o, q, s) before and (b, d, f, h, j, l, n, p, r, t) after bedding correction. Solid (square)/dashed (triangle) symbols represent downward/upward inclinations; shallow symbols indicate the overall-mean directions with 95% confidence limits.

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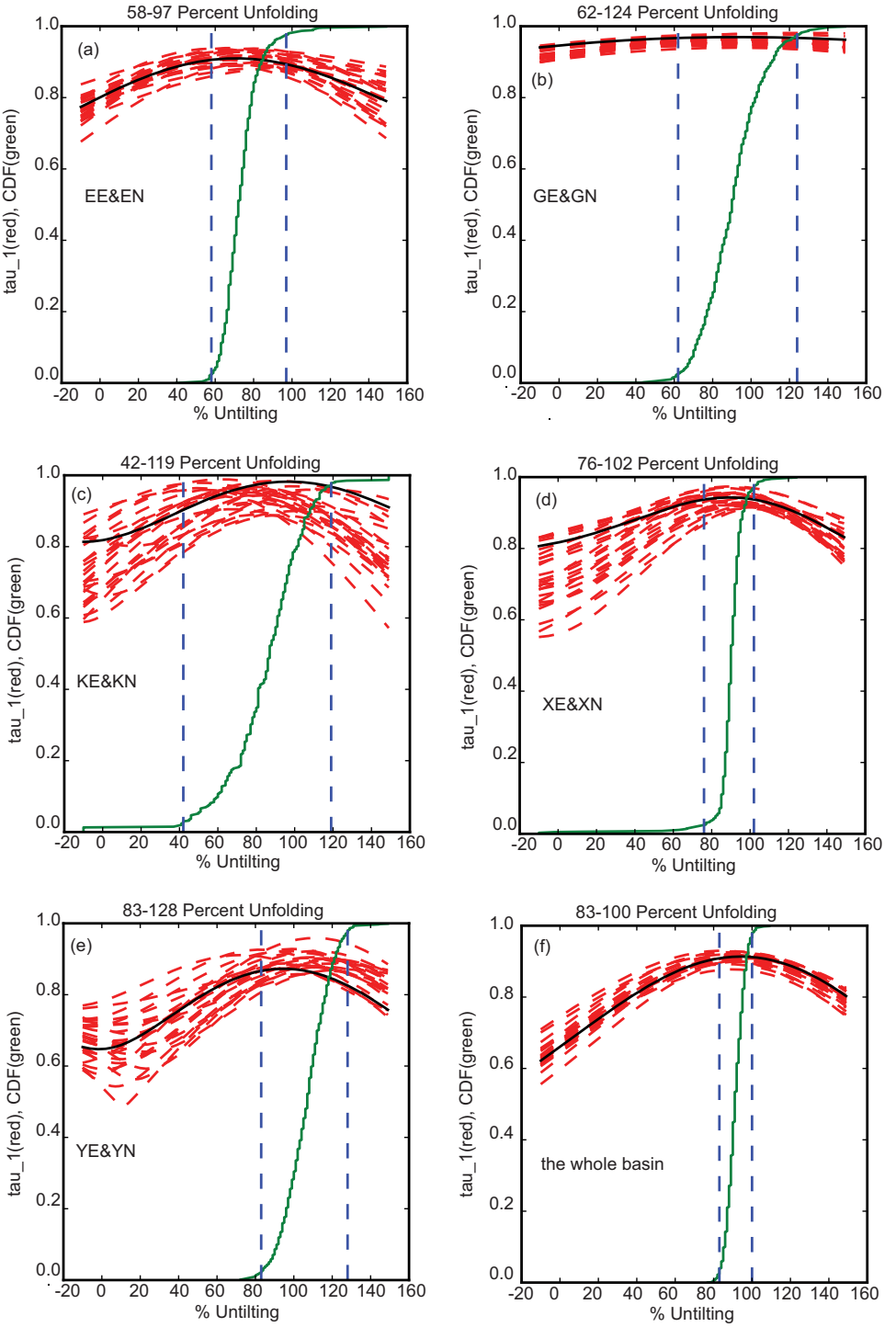


Figure 5. Parametric progressive fold test from the output of the PmagPy software (2.206) [*Tauxe*, 2010]. Dashed red curves show the variation of the principal eigenvalue of the orientation matrix (τ_1) with percent unfolding. Green curve shows the cumulative distribution function for 1000 para-data sets. (b, c, d, e, f) Indicates positive fold tests, because the region of 95% confidence for the percent unfolding that maximizes τ_1 (vertical dashed lines) contains the 100% unfolding value. (a) An undetermined fold test, because the region of 95% confidence for the percent unfolding that maximizes τ_1 contains neither 100% nor 0% unfolding value. The whole basin results exclude the Gahai section (f).

4.3. Eboliang

Eboliang is a butterfly-shaped anticline situated ~30 km to the south of the ATF. The Xiaganchaigou Formation (EE) outcrops along a hanging wall of an east-vergent, north-south trending thrust fault. Collected samples are light brownish red, dark brownish red, light brownish yellow, light yellow, and light cyan mudstone

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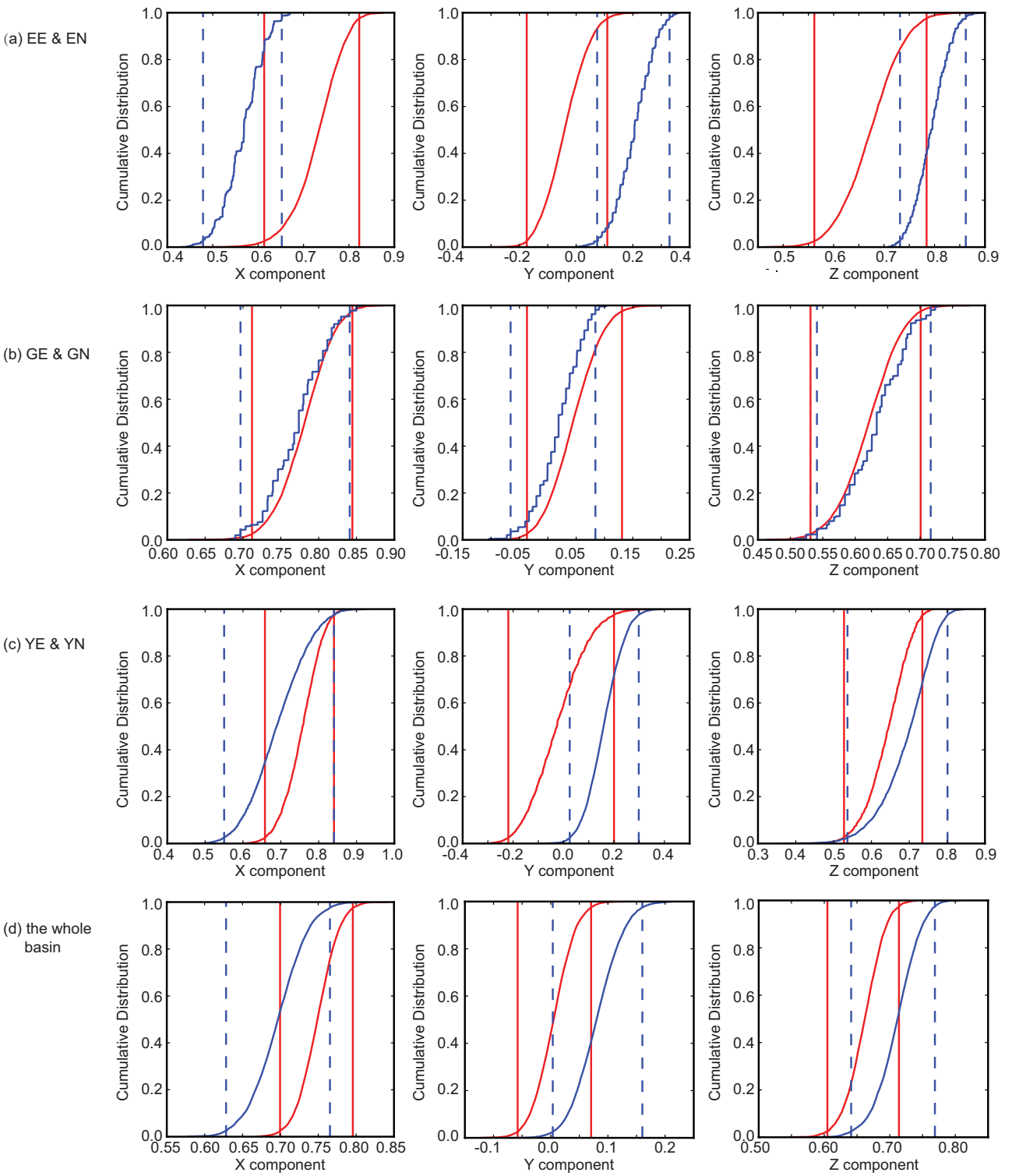


Figure 6. Parametric reversal test from the output of PmagPy software (2.206) [*Tauxe*, 2010]. Cumulative distribution and 95% confidence intervals for Cartesian magnetization components X, Y, Z are displayed for normal (red) and reversed (blue) polarity directions. Overlap of 95% confidence intervals for normal and reversed directions on all three components (shaded regions) indicates positive reversal tests for all cases. The whole basin results exclude the Gahai section (d).

with a little earth yellow siltstone. The Xiayoushashan Formation (EN) samples collected at the east limb and south plunging crown of the Eboliang anticline are dark brownish red, yellow and orange sandstone, coarse sandstone, and sandy conglomerate, with dark yellow mudstone.

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For samples from the eight sites of the Xiaganchaigou Formation (EE), apart from a LTC, most thermal demagnetizations reveal one HTC with unblocking temperatures dominantly around 500–590°C (Figure 3b), so the stable component of magnetization is thought to be carried by magnetite. The eight sites from the Xiaganchaigou Formation (EE) provide a site-mean direction for the HTCs of D = 22.3°, I = 51.3° before and of D = 6.9°, I = 55.1° after tilt-correction (Figures 4c and 4d). Apart from a LTC, most thermal demagnetizations reveal one HTC for the samples from the Xiayoushashan Formation (EN), with the temperatures between 150–450°C and 650–690°C (Figure 3j), or sometimes between 150–400°C and 500–600°C, indicating that the stable components of magnetization are carried by magnetite and hematite for different specimens. The eight sites from the Xiayoushashan Formation (EN) provide a site-mean direction of D = 333.8°, I = 36.2° before and of D = 0.9°, I = 37.3° after tilt correction (Figures 4e and 4f). Noticeable, for both the specimens from the Xiaganchaigou Formation and the Xiayoushashan Formation, J_r-T curves see a dramatic decrease at temperature <150°C, indicating the existent of a large amount of goethite.

To assess the timing of magnetization to tilting of the two formations (EE & EN), a combined fold test was performed. The 16 site-mean directions from these two formations provide a positive fold-test result ($k_s/k_g = 1.99$, $F(30,30)_{0.05} = 1.84$) [*McElhinny*, 1964]. Although the parametric progressive fold test is intermediate, the 95% confidence bounds on the degree of untilting required for maximizing concentration of the data τ_1 of orientation matrix are close to include 100% (Figure 5a) [*Tauxe*, 2010]. The site-mean HTC directions pass the bootstrap reversal test (Figure 6a) [*Tauxe*, 2010].

4.4. Heishiqiu

Heishiqiu is an extension of the Saishiteng Shan, located at the northwesternmost end of the northern Qaidam basin, 20 km west to Lenghu Town and 10 km north to Xiaosaishiteng Shan. It is constituted of a south-vergent monocline and the oldest outcrop stratum is the Lulehe Formation. The Xiaganchaigou Formation (LE) samples are dark red, brick red, pink, and orange sandstone and sandy conglomerate, while the Xiayoushashan Formation (LN) samples are different shades of yellow mudstone with a little sandstone.

Two typical thermal demagnetization paths are shown in Figures 3c and 3g for the Xiaganchaigou Formation (LE) and the Xiayoushashan Formation (LN). A LTC is resolved between NRM-100 and 200–250°C, and a HTC is isolated between 200–500°C and 500–590°C (Figure 3c), or sometimes higher temperatures up to around 680°C (Figure 3g). The overall site-mean HTC direction (LE and LN) is of D = 188.3°, I = −6.4° before and of D = 178.1° and I = −47.1° (Figures 4o and 4p) after tilt correction for these two formations at Heishiqiu. The fold test is inconclusive due to small variation in bedding attitudes, and there are not enough normal polarity results to do reversal test.

4.5. Lulehe

Lulehe is at the natural connection between Saishiteng Shan and Lvliang Shan. The NW-SE-trending Lulehe anticline is thought to be the most complete Cenozoic profile in the northern Qaidam basin, outcropping Jurassic to Quaternary from the core to the two limbs. The Xiaganchaigou Formation (YE) collected from both two limbs are brownish yellow and brownish red mudstone and orange siltstone with a little sandstone, and light brown muddy siltstone with a little brown sandstone. The Xiayoushashan Formation (YN) in the northeast limb is seriously covered by the Quaternary sediments, so all samples were collected in the southwest limb, consisted of grass green and light grey sandstone, coarse sandstone, sandy conglomerate, and orange and red mudstone.

Two components are identified by progressively thermal demagnetization. LTC was generally removed in temperature range of NRM-100°C to 200−300°C. A HTC was revealed generally between 250–500°C and ~680°C (Figure 3k) for most specimens from the Xiaganchaigou Formation (YE), while it was mainly revealed between 250–450°C and around 600°C (Figure 3e) for most specimens from the Xiayoushashan Formation (YN). For the specimens from the Xiaganchaigou Formation (YE), the site-mean direction is determined for HTC components from eight sites. The HTC components yield a site-mean direction of D = 357.4°, I = 26.2° before and of D = 10.8°, I = 40.5° after tilt correction (Figures 4q and 4r). For the specimens from the Xiayoushashan Formation (YN), the site-mean direction is determined for HTC components from seven out of eight sites collected. The HTC components produce a site-mean direction of D = 16.3°, I = −0.1° before and of D = 0.0°, I = 45.0° after tilt correction (Figures 4s and 4t).

The grouping of the dataset from the Xiaganchaigou and Xiayoushashan formations (YE & YN) increases obviously after tilt correction with the ratio $k_s/k_g = 5.32$ ($F(28,28)_{0.05} = 1.88$), defining a positive fold test

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[*McElhinny*, 1964]. Because the 95% confidence bounds on the degree of untilting required for maximizing concentration of the data τ_1 of orientation matrix are 83% and 128% (Figure 5e), the parametric progressive fold test indicates a prefold origin [*Tauxe*, 2010]. Parametric reversal test is also positive for the two formations at Lulehe (Figure 6c) [*Tauxe*, 2010].

4.6. Kushuiquan

Kushuiquan, located in the middle of the northern Qaidam basin and 10 km to the south of Lvliang Shan, is a natural westward extension of Xitie Shan. Similar to Lulehe, Kushuiquan is also an NW-SE trenching anticline, outcropping Cretaceous in the core. The Xiaganchaigou Formation (KE) samples collected in the south limb of Kushuiquan are dark red mudstone, silty mudstone, siltstone and sandstone with light grey, and yellowish green sandstone. The specimens for the Xiayoushashan Formation (KN) collected at the connection of the Kushuiquan anticline and Xitie Shan, are generally white, yellowish green, light grey, and yellow sandstone.

For the specimens from the Xiaganchaigou Formation (KE), three paleomagnetic components were observed: (1) a LTC isolated from temperatures between NRM-100 and 100–300°C, (2) an intermediate temperature component (ITC) calculated temperatures mainly between 100–300 and 300–500°C, and (3) a HTC generally separated from temperatures between 300–500 and 580–690°C (Figure 3l). Site-mean directions were calculated for LTCs from eight sites, for ITCs from five sites and for HTCs from three sites. A postfold origin for LTC is suggested by a negative fold test result. Moreover, the ITC ($D_g = 4.8°$, $I_g = 48.8°$) is close to LTC ($D_g = 3.6°$, $I_g = 57.6°$) in geographic coordinates, closing to the present geomagnetic field direction computed from the International Geomagnetic Reference Field (D = 0°, I = 57°), while HTC ($D_g = 0.6°$, $I_g = -2.2°$; $D_s = 3.2°$, $I_s = 39.3°$) is totally different from LTC. It implies that LTC and ITC are both postfold origin, and HTC is prefold origin. For the specimens from the Xiayoushashan Formation (KN), two paleomagnetic components were observed: (1) a LTC separated from temperatures around 100–250°C, (2) a HTC with unblocking temperature mainly <600°C (Figure 3i).

The HTC components for both the Xiaganchaigou and Xiayoushashan formations (KE & KN) yield a site-mean direction of D = 359.9°, I = 30.9° before and of D = 8.7°, I = 32.9° after tilt correction (Figure 4 (k, l)). The parametric progressive fold test indicates a pre-fold origin (Figure 5c) [*Tauxe*, 2010], and improved clustering of site-mean directions upon restoring local bedding to horizontal compared with in situ direction also supports that the remanence was acquired prior to folding ($K_s/K_g = 4.46$, $F(10,10)_{0.05} = 2.97$) [*McElhinny*, 1964].

4.7. Gahai

Gahai is a southwest-vergent arcuate structure, situated at easternmost end of the northern Qaidam basin, ~40 km north to Delingha City and ~20 km north to Gahai. The samples from the Xiaganchaigou Formation (BE) are brownish sandstone, coarse sandstone and sandy conglomerate, while the samples from the Xiayoushashan Formation (BN) are roughly different shades of orange silty mudstone and muddy siltstone.

For specimens from the Xiaganchaigou Formation, erratic demagnetization paths prevent determination of HTC directions. These sites are discarded from further HTC analysis. For the Xiayoushashan Formation (BN), thermal demagnetizations were generally conducted up to a maximum temperature of 690°C. As shown by the J_r-T curves, the most important decrease of nature remanent magnetization occurred around 620–680°C, indicating that various phase of hematite is the major magnetic carrier. In addition, a minor decrease took place around 500–590°C, revealing the existent of magnetite. Apart from a LTC, most thermal demagnetizations reveal only one high temperature characteristic remnant magnetization for the Xiayoushashan Formation, mainly unblocked by temperatures around 680°C (Figure 3a). The five sites provide a site-mean direction for the HTC of D = 213.2°, I = −18.8° before and of D = 207.5°, I = −31.4° after tilt correction with slight improvement of data grouping following tilt correction with $k_s/k_g = 1.35$ (Figures 4a and 4b). The *Watson and Enkin* [1993] fold test defines an optimal concentration at 100% unfolding with 95% uncertainties ranging from 57% to 136%, indicating a prefold origin.

5. Discussion

5.1. Regional Summary Results

A basin-scale LTC has been resolved in 488 specimens of the Xiaganchaigou and Xiayoushashan formations. The overall sample-mean direction is D = 354.9°, I = 58.0°, k = 18.1, $a_{95} = 1.5$ before tilt-correction and D = 344.8°, I = 63.7°, k = 3.4, $a_{95} = 4.1$ after tilt-correction (Figure 7). The high k_g/k_s value (5.32) suggests a

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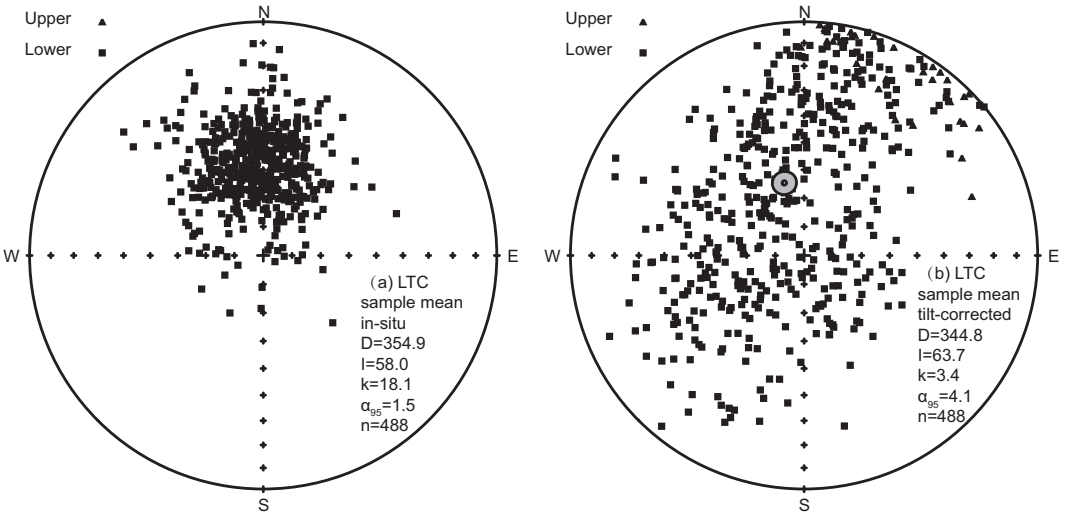


Figure 7. Equal-area projections of sample-mean low-temperature component (LTC) results from the whole basin (a) before and (b) after bedding correction. Shallow symbols indicates the overall-mean directions with 95% confidence limits. Square/triangle symbols represent downward/upward inclination.

postfold origin. Moreover, the in situ LTC direction is close to the present geomagnetic field direction computed from the International Geomagnetic Reference Field of the Qaidam basin (D = 0°, I = 58°).

The HTC directions from different sections inside the Qaidam basin are similar to each other, except those from the easternmost Gahai section (Figures 8a and 8b). To verify if the Qaidam basin experienced a basin-scale rotation during the Cenozoic and the timing of magnetization with respect to tilting, we performed a fold test using the HTC directions throughout the basin (excluding results from Gahai). The 67 sites from the Xiaganchaigou and Xiayoushashan formations provide a site-mean HTC direction of D = 359.3°, I = 31.4°, k = 4.8, a_{95} = 8.9 before and D = 2.5°, I = 42.9°, k = 20.3, a_{95} = 3.9 after tilt correction (Figures 8c and 8d). The parametric progressive fold test suggests a prefold origin (Figure 5f) [*Tauxe*, 2010], and the parametric reversal test also indicates a positive result (Figure 6d) [*Tauxe*, 2010].

The dual polarities and positive fold tests suggest that the high temperature characteristic remnant magnetization (ChRM) directions were acquired at or close to the time of deposition. Therefore, we interpret the ChRM directions as primary.

5.2. Sample-Mean Results

To verify the reliability of our site-mean results, sample-mean results are also calculated (Table 1). The sample-mean results are similar to site-mean results. *Deenen et al*. [2011] define a range of A95 values that are consistent with directional scatter resulting from secular variation of the magnetic field. Most A95 values of sample-mean results fall into the range between $A95_{min}$ and $A95_{max}$, except the Xiaganchaigou Formation from Lulehe (YE) and the whole basin (E_3), of which A95 is a little higher than $A95_{max}$. The results imply that most observed scatters can be explained by secular variation, also indicating that the sediments have adequately recorded the geomagnetic field at the time of deposition. Therefore, we conclude that our paleomagnetic results are suitable for testing the rotation of the Qaidam basin.

5.3. No Basin-Scale Rotation of the Qaidam Basin

Except at Gahai, most paleomagnetic results are comparative with the paleomagnetic field directions predicted in the Qaidam basin (38.1°N, 93.4°E) from the mean Eurasian APW path [*Besse and Coutillot*, 2002] (Figure 9). The comparative directions are D/I = 11.2/58.4° at 35 Ma and D/I = 6.7/59.8° at 15 Ma. Therefore, our paleomagnetic results show that the Qaidam basin did not rotate with respect to Eurasia since the Middle to Late Eocene. We observe a 19.0 ± 7.9° postEarly Miocene clockwise rotation at Gahai [*Besse and Coutillot*, 2002]. The rotation appears to be restricted to the Gahai region.

Dupont-Nivet et al. [2002] demonstrated that there has been no Neogene vertical axis rotation of the Qaidam basin with paleomagnetic data from Eboliang and Xiao Qaidam. Our results build substantially on this earlier

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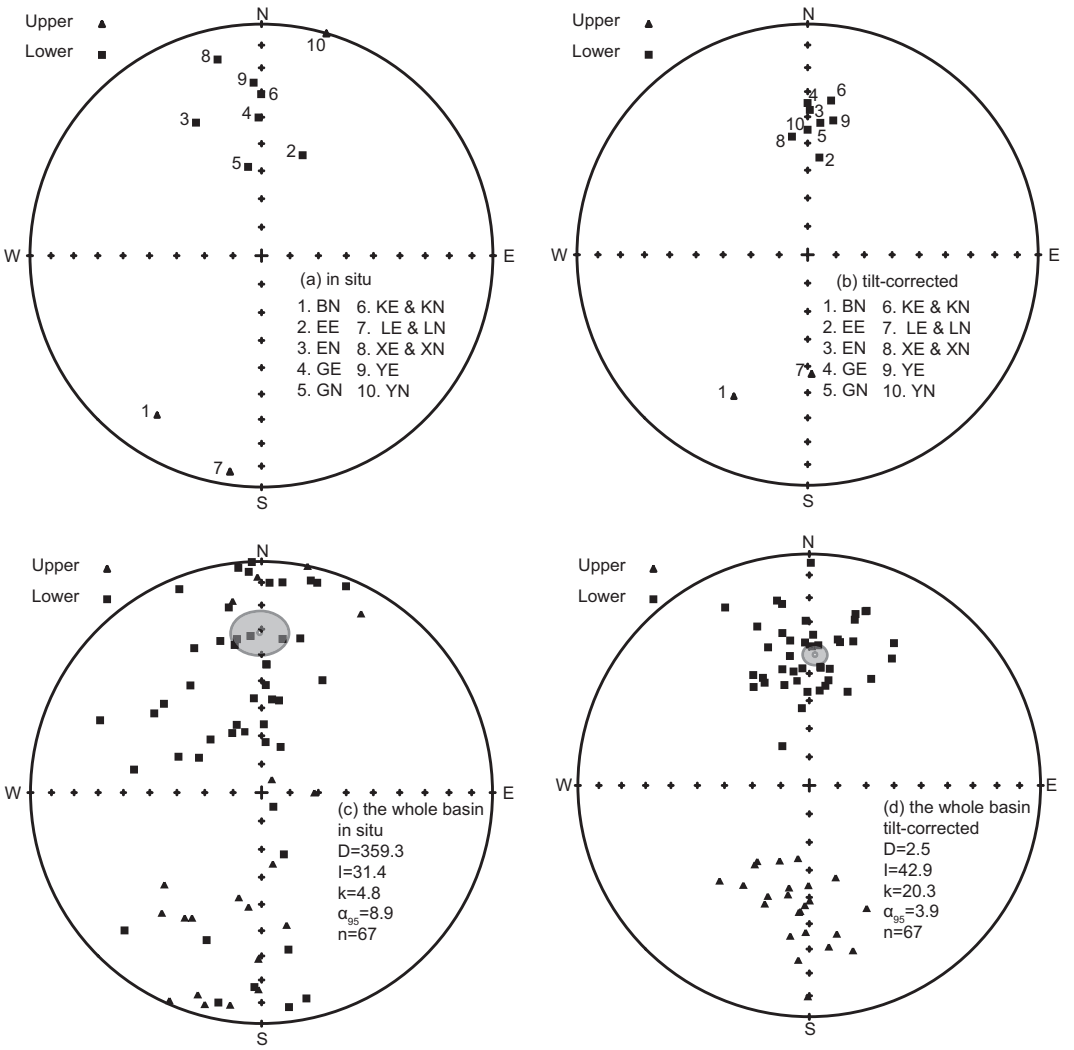


Figure 8. Equal-area projections of a sum of site-mean HTC directions of different sections (a) before and (b) after bedding correction and equal-area projections of the site-mean HTC directions of the whole basin (c) before and (d) after bedding correction. Shallow symbols indicate the overall-mean directions with 95% confidence limits. Square/triangle symbols represent downward/upward inclination. The whole basin results exclude the Gahai section (c, d).

study and confirm those initial interpretations. Moreover, paleomagnetic data from Miocene rocks at four localities indicates that the Nan Shan fold-and-thrust belt did not experience whole-scale vertical axial rotation since the early Neogene [*Dupont-Nivet et al.*, 2003]. North of the Nan Shan, the Hexi corridor has not undergone significant and systematic tectonic vertical axial rotation since at least the early Cretaceous [*Dupont-Nivet et al.*, 2003]. To the southwest, the observed declination at Tula, which is far less than the difference predicted for oroclinal bending, suggests that crustal displacement between the Tarim Basin and the Qaidam basin is accommodated by strike-slip motion on the Altyn Tagh Fault rather than distributed shear within the northern Tibetan Plateau [*Dupont-Nivet et al.*, 2004b]. *Sun et al.* [2006] also concluded that the Qaidam Block had not experienced a large-scale vertical axial rotation since the Early Cretaceous with respect to the North China Block (NCB) pole, on the basis of the existing regional paleomagnetic data. However, *Chen et al.* [2002] proposed that the Qaidam basin experienced a clockwise rotation between 24 and 10 Ma. Nevertheless, that study found no Cenozoic rotation at Huatugou (consistent with our results from the same region), and only a 16.5 ± 11.7° rotation at Mahai with respect to the Besse and Coutillot pole [*Besse and Coutillot*, 2002; *Dupont-Nivet et al.*, 2003]. With the context of these previous paleomagnetic data and our new paleomagnetic results, we propose that the Qaidam basin did not undergo basin-scale vertical axis rotation since at least the Middle Eocene. Local clockwise rotation took place at a few special locations along northern margin of the Qaidam basin, such as Mahai and Gahai, probably

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Table 1. Site-Mean and Sample-Mean Paleomagnetic Results in the Qaidam Basin

Name	Method	N	D_g	I_g	k_g	a_{95g}	D_s	I_s	k_s	a_{95s}	k_s/k_g	K	A95	$A95_{max}$	$A95_{min}$
BN	Site	5	213.2	−18.8	34.7	13.2	207.5	−31.4	47.0	11.3	1.35	58.1	10.1	29.7	8.9
	Sample	39	213.6	−18.1	16.8	5.8	208.3	−31.1	18.6	5.5	1.11	24.9	4.7	8.2	3.9
EE	Site	8	22.3	51.3	16.8	13.9	6.9	55.1	21.4	12.3	1.27	12.9	16.0	22.1	7.4
	Sample	55	22.1	52.8	14.5	5.2	7.1	55.5	16.2	4.9	1.12	10.6	6.2	6.6	3.4
EN	Site	8	333.8	36.2	10.6	17.8	0.9	37.3	17.0	13.8	1.60	16.9	13.9	22.1	7.4
	Sample	64	334.8	35.8	9.0	6.3	359.9	37.5	12.2	5.3	1.36	12.2	5.3	6.0	3.2
GE	Site	7	358.8	40.1	26.0	12.1	360.0	34.6	36.1	10.2	1.39	47.4	8.9	24.1	7.8
	Sample	55	358.1	40.0	17.8	4.7	359.9	35.0	21.2	4.3	1.19	22.9	4.1	6.6	3.4
GN	Site	8	351.4	58.2	41.2	8.7	5.5	42.1	56.6	7.4	1.37	65.3	6.9	22.1	7.4
	Sample	56	350.1	58.6	26.1	3.8	4.8	42.5	31.2	3.5	1.20	30.2	3.5	6.5	3.4
KE and KN	Site	6	359.9	30.9	3.7	40.6	8.7	32.9	16.5	17.0	4.46	26.1	13.4	26.5	8.3
	Sample	40	358.3	28.8	3.3	14.9	6.8	35.6	10.2	7.5	3.09	9.9	7.6	8.0	3.9
LE and LN	Site	6	188.3	−6.4	44.3	10.2	178.1	−47.1	38.8	10.9	0.88	40.8	10.6	26.5	8.3
	Sample	46	187.7	−6.5	17.1	5.2	179.1	−46.4	16.2	5.4	0.95	14.0	5.8	7.3	3.7
XE and XN	Site	9	347.4	14.3	4.7	26.4	352.5	47.2	24.9	10.5	5.30	23.5	10.8	20.5	7.1
	Sample	64	347.6	15.6	4.6	9.3	352.8	47.9	16.2	4.5	3.52	13.7	5.0	6.0	3.2
YE	Site	8	357.4	26.2	1.6	68.9	10.8	40.5	12.0	16.6	7.50	13.0	15.9	22.1	7.4
	Sample	45	350.0	31.1	1.7	24.6	8.8	47.8	9.9	7.1	5.82	8.6	7.7	7.5	3.7
YN	Site	7	16.3	−0.1	19.8	13.9	0.0	45.0	19.2	14.1	0.97	14.1	16.6	24.1	7.8
	Sample	37	15.7	0.3	15.5	6.2	360.0	44.3	15.5	6.2	1.00	12.9	6.8	8.4	4.0
E_3	Site	32	6.1	32.2	4.5	13.5	4.4	44.7	18.7	6.0	4.16	18.5	6.1	9.2	4.3
	Sample	222	4.5	33.5	4.4	5.1	3.2	46.0	12.2	2.8	2.77	10.6	3.0	2.7	2.0
N_2^1	Site	40	359.5	29.5	5.0	11.3	4.6	40.3	19.5	5.2	3.90	19.5	5.3	8.0	3.9
	Sample	279	358.4	30.8	4.7	4.4	4.7	40.7	13.9	2.3	2.96	13.5	2.4	2.4	1.8

D_g, I_g (D_s, I_s), declination, and inclination of direction in in situ coordinate (tilt-corrected coordinates); α_{95g}(α_{95s}), the radius that the mean direction lies within 95% confidence in in situ coordinate (tilt-corrected coordinates); k_g (k_s), the best estimate of precision parameter in in situ coordinate (tilt-corrected coordinates); and K, precision parameter determined from the mean virtual geomagnetic pole directions (VGPs). A95, cone of confidence determined from the mean VGP direction; $A95_{min}$ ($A95_{max}$), minimum (maximum) value of the A95 for the given data set based on *Deenen et al.* [2011]. Remagnetization circles are joined the site-mean calculation but not joined sample-mean calculation.

related to the thrusting and folding [*Bates*, 1989; *Allerton*, 1998]. The thrusting and folding is generally thought to be a consequence of slip from the ATF being partitioned into the Qilian Shan-Nan Shan [*Meyer et al.*, 1998; *Yin and Harrison*, 2000; *Darby et al.*, 2005; *Zhuang et al.*, 2011]. The interaction between the sedimentary rock and bedrock along the northern boundary of the Qaidam basin also implies that the rigid Qaidam basement may have been broken at these locations.

5.4. Shallow Inclinations

Although most HTC results show no obvious difference between the observed and predicted declinations, the observed inclinations are significantly shallower than the predicted directions. Therefore, the inclination-shallowing effect should not be ignored at these sampling sections.

A shallow bias in paleomagnetic inclinations in Central Asia recorded in clastic sedimentary rocks has been noted and studied for decades [*Westphal*, 1993; *Cogné et al.*, 1999; *Si and Van der Voo*, 2001; *Tauxe and Kent*, 2004]. The E/I correction based on the statistical paleosecular variation models of the geocentric axial dipole magnetic field [*Tauxe and Kent*, 2004] was applied to the Xiaganchaigou Formation and the Xiayoushashan Formation in the Qaidam basin, respectively (Figure 9). Overall, the shallowing amount in the Xiaganchaigou Formation is obviously lower than that of the Xiayoushashan Formation, and the distribution of sample-mean directions of the Xiaganchaigou Formation is closer to the scatter predicted from statistical models of paleosecular variation than that of the Xiayoushashan Formation. The sample-mean inclination of the Xiaganchaigou Formation is corrected from 46.0 ± 2.8° to 51.7° with 95% confidence limit between 46.4° and 57.7°. The sample-mean inclination of the Xiayoushashan Formation is corrected from 40.7 ± 2.3° to 61.4° with 95% confidence limit between 54.4° and 67.0°. These corrected inclinations are statistically indistinguishable from the paleomagnetic field directions predicted in the Qaidam basin with respect to the mean Eurasian APW path [*Besse and Coutillot*, 2002]. It indicates that the paleographic site of the Qaidam basin did not move north or south in the Cenozoic by an amount that can be detected paleomagentically.

5.5. Implication for Deformation Behavior of a Quasi-Rigid Qaidam Block During the Indo-Asian Collision

Compared to strongly deformed surrounding areas, the Cretaceous to Tertiary sedimentary of the basin is only deformed by broad, open fold [*Meyer et al.*, 1998; *Zhou et al.*, 2006; *Yin et al.*, 2007, 2008a, 2008b]. This

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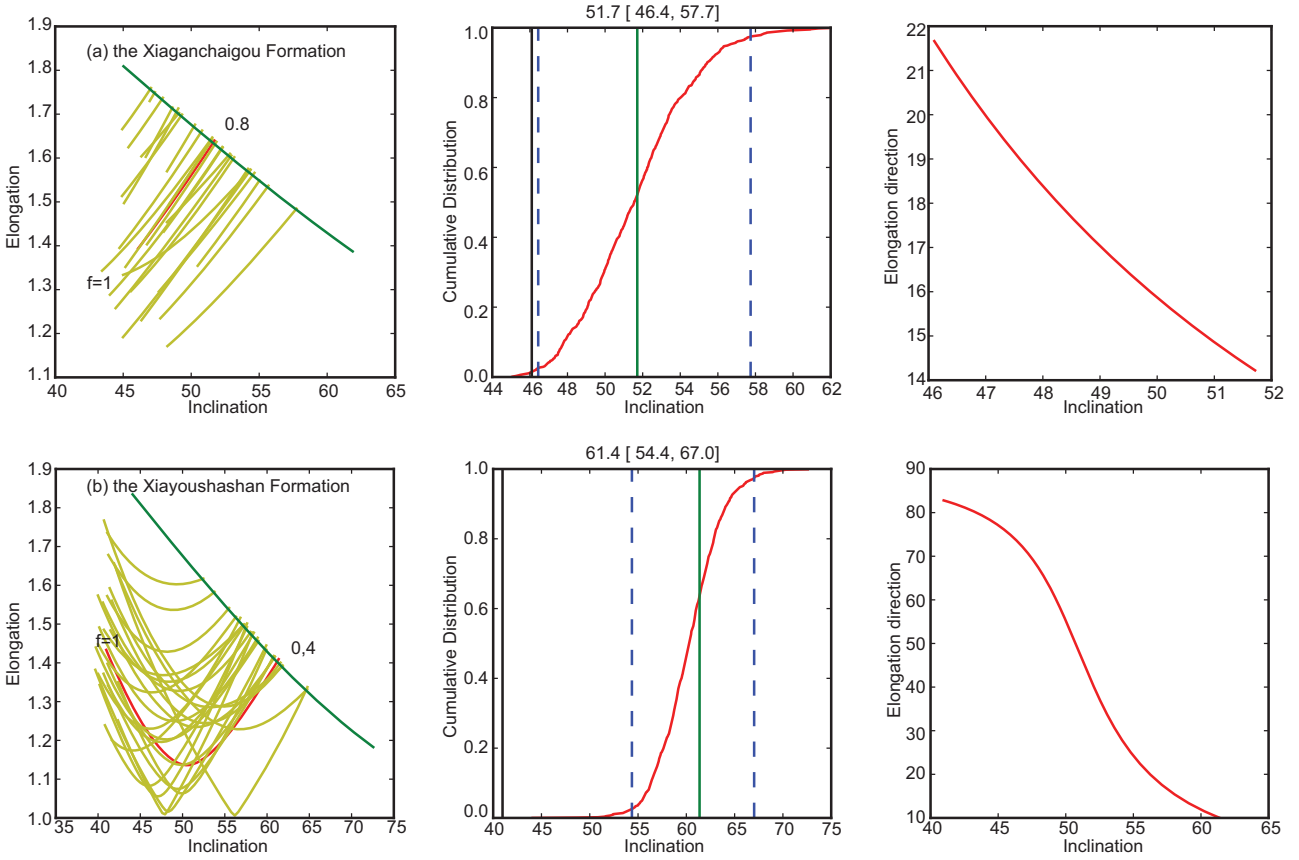


Figure 9. Correction for the inclination shallowing using the E/I method [*Tauxe and Kent*, 2004]. The directions were "unflattened" by inverting the inclination error formula [*King*, 1955]: $\tan I_{cor} = (1/fest) \tan I_{obs}$ where I_{cor} is the corrected inclination, fest is the estimated flattening factor, and I_{obs} is the measured inclination. The PmagPy software package (2.206) of *Tauxe* [2010] was used to find the optimum flattening factor which yields an elongation of the distribution of "unflattened" directions most consistent with the TK03.GAD paleosecular variation model of *Tauxe and Kent* [2004]. The cumulative distribution shows the 95% confidence limit of the corrected inclinations.

observation suggests that the Qaidam basement may be stronger than basement in the surrounding regions, which may lead to a change in the elevation and intensity of deformation between the basin and its surrounding area [*Zhu et al*., 1995; *Cook and Royden*, 2008; *Dayem et al*., 2009]. The average P-wave velocity of the crust and upper mantle under the Qaidam basin is also faster than that under the Tibetan Plateau [*Zhu and Helmberger*, 1998], consistent with a cold and strong crust. Although the average crustal thickness of the Tibetan Plateau is much higher than that of the Qaidam basin, the effective elastic thickness of the Qaidam basin is apparently higher than that of surrounding regions [*Jordan and Watts*, 2005]. The effective elastic thickness of the Qaidam basin is estimated to be 50–60 km, similar to the Sichuan basin, while that of the Tibetan Plateau is estimated to be $<$20 km [*Jordan and Watts*, 2005]. Except at the easternmost Gahai section, our paleomagnetic results are very similar in magnitude and direction at different localities in the whole basin (Figure 10), which also supports a relatively rigid Qaidam block.

The effective elastic thickness is generally lower in the surrounding areas of the Qaidam basin, including the Eastern Kunlun Mountains and the Qilian Mountains [*Jordan and Watts*, 2005]. However, the effective elastic thickness of the Altyn Tagh Mountains is relatively high [*Jordan and Watts*, 2005], suggesting that the Qaidam Block and the Tarim Block are in rigid contact, totally different from the other boundaries of the Qaidam basin. Thus, although the contact between the Qaidam and Tarim blocks is under N-S compression from India-Asian collision, vertical axis rotation is limited and strike-slip movement along the ATF is produced. Most shortening is absorbed by the surrounding areas of the Qaidam basin, such as the Qilian Shan-Nan Shan Trust Zone.

5.6. Implication for Crustal Movement on Either Side of the Kunlun Fault

Clockwise rotation of the Qaidam basin is predicted in many tectonic models because it is bounded by two large left-lateral strike-slip faults: the ATF and the KLF [*Engl et al*., 1986; *Avouac and Tapponnier*, 1993].

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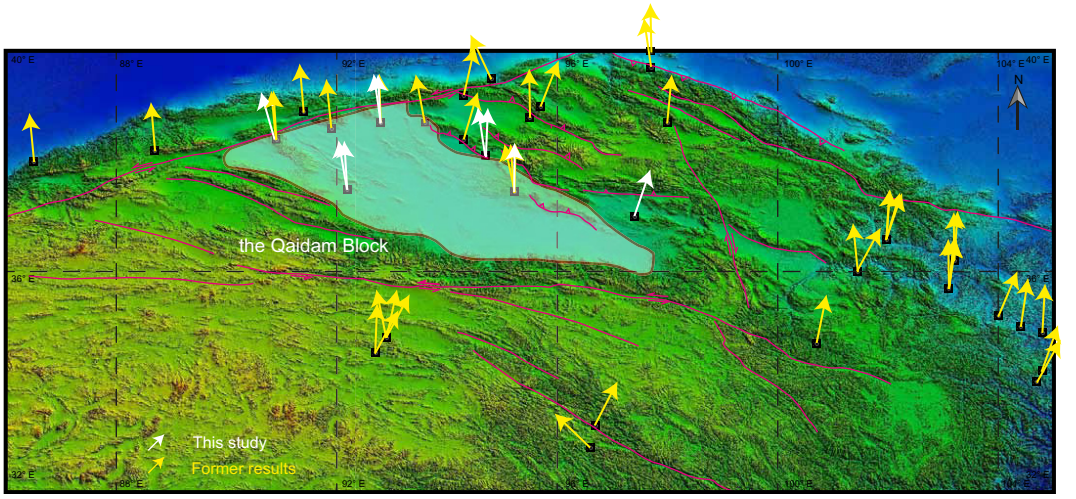


Figure 10. Vertical axis rotations with respect to the Besse and Courtillot poles of Eurasia [*Besse and Courtillot*, 2002] in the northern Tibetan Plateau (Data from this study, *Dupont-Nivet et al.* [2002, 2003, 2004a], *Chen et al.* [2002], *Flynn et al.* [1999], *Fang et al.* [2003], *Cogné et al.* [1999], *Rumelhart et al.* [1999], *Yan et al.*[2006], *Lu et al.* [2009], *Sun et al.* [2005], *Liu et al.* [2003]; see supporting information Table S3).

Present GPS measurement and geological evidence all suggest a slip rate of ~10 mm/yr along the Kunlun Fault [*Zhang et al.*, 2004; *Fu et al.*, 2005; *Fu and Awata*, 2007; *van der Woerd*, 1998; *Wang et al.*, 2001; *Li et al.*, 2005; *Lin and Guo*, 2008]. Recent results on slip rates along AFT are similar at ~10 mm/yr [*Yin et al.*, 2002; *Shen et al.*, 2001; *Bendick et al.*, 2000; *Wallace et al.*, 2004; *Zhang et al.*, 2007; *Elliott et al.*, 2008; *Cowgill*, 2007; *Cowgill et al.*, 2009; *Duvall and Clark*, 2010; *Yuan et al.*, 2013]. The paleomagnetic results presented above imply that most clockwise rotation occurred south to the KLT [*Halim et al.*, 1998; *Cogné et al.*, 1999; *Liu et al.*, 2003] and east to the Qaidam basin [*Yan et al.*, 2006; *Dupont-Nivet et al.*, 2004a, 2008], in the other words, outside of the Qaidam basin. Given that the ATF moved northeastward on its south side and the KLF moved westward on its north side, we would predict a clockwise rotation of the Qaidam block (Figure 11c). However, our own and previous paleomagnetic results show no integral rotation of the Qaidam basin. Notably, geodetic measurements indicate that the north side of the KLF moves toward northeast [*Zhang et al.*, 2004; *Gan et al.*, 2007; *Duvall and Clark*, 2010; *Zheng et al.*, 2013]. And although the KLF and the ATF move simultaneously, there is no evidence that the ATF is cut by the KLF.

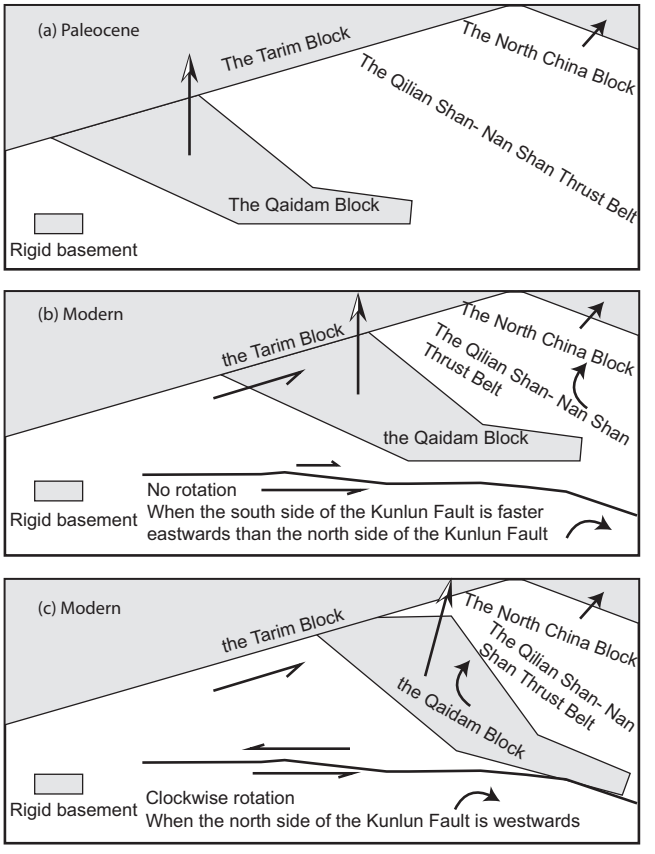


Figure 11. Simple tectonic evolution model of the Qaidam basin. (a) Paleocene; (b) Present when crust on the south side of the Kunlun Fault moves eastward faster than crust on the north side; (c) Present when crust on the north side of the Kunlun Fault moves westward. Our paleomagnetic results support the model shown in Figure 11b.

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These observations suggest that crust north of the KLF does not move westward on its north side in an absolute sense [*van Hinsbergen et al*., 2011]. Therefore, an eastward-movement of crust, which is much faster on the south side than that on the north side of the KLF, leads to no rotation of the Qaidam basin since the Eocene (Figure 11b).

6. Conclusion

Eocene and Miocene sedimentary units within the Qaidam basin preserve primary detrital remanent magnetizations that can be used for tectonic studies of the region. Our paleomagnetic results, collected from outcrops throughout the Qaidam basin, indicate that the basin has undergone no large systematic vertical axis rotation since at least the Eocene. Clockwise rotations occurred at only a few specific places along the northern margin of the basin, likely related to localized folding and thrusting. The regional summary of paleomagnetic data, together with effective elastic thickness and other geophysical evidence, suggests that the Qaidam basin is a relatively rigid block in the Tibetan Plateau.

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References

Allerton, S. (1998), Geometry and kinematics of vertical-axis rotations in fold and thrust belts, *Tectonophysics*, *299*, 15–30, doi:10.1016/S0040-1951(98)00196-6.

Avouac, J. P., and P. Tapponnier (1993), Kinematic model of active deformation in central Asia, *Geophys. Res. Lett.*, *20*(10), 895–898, doi:10.1029/93GL00128.

Bates, M. P. (1989), Palaeomagnetic evidence for rotations and deformation in the Nogueras Zone, Central Southern Pyrenees, Spain, *J. Geol. Soc. London*, *146*, 459–476, doi:10.1144/gsjgs.146.3.0459.

Bendick, R., R. Bilham, J. Freymueller, K. Larson, and G. Yin (2000), Geodetic evidence for a low slip rate in the Altyn Tagh fault system, *Nature*, *404*, 69–72, doi:10.1038/35003555.

Besse, J., and V. Courtillot (2002), Apparent and true polar wander and the geometry of the geomagnetic field over the last 200 Myr, *J. Geophys. Res.*, *107*(B11), 2300, doi:10.1029/2000JB000050.

Chang, H., H. Ao, Z. S. An, X. M. Fang, Y. G. Song, and X. K. Qiang (2012), Magnetostratigraphy of the Suerkuli Basin indicates Pliocene (3.2 Ma) activity of the middle Altyn Tagh Fault, northern Tibetan Plateau, *J. Asian Earth Sci.*, *44*, 169–175, doi:10.1016/j.jseaes.2011.10.013.

Chen, Y., S. Gilder, N. Halim, J. P. Cogné, and V. Courtillot (2002), New paleomagnetic constraints on central Asian kinematics: Displacement along the Altyn Tagh fault and rotation of the Qaidam Basin, *Tectonics*, *21*(5), 1042, doi:10.1029/2001TC901030.

Cogné, J. P., N. Halim, Y. Chen, and V. Courtillot (1999), Resolving the problem of shallow magnetizations of Tertiary age in Asia: Insights from paleomagnetic data from the Qiangtang, Kunlun, and Qaidam blocks (Tibet, China), and a new hypothesis, *J. Geophys. Res.*, *104*, 17,715–17,734, doi:10.1029/1999JB900153.

Cook, K. L., and L. H. Royden (2008), The role of crustal strength variations in shaping orogenic plateaus, with application to Tibet, *J. Geophys. Res.*, *113*, B08407, doi:10.1029/2007JB005457.

Cowgill, E. (2007), Impact of riser reconstructions on estimation of secular variation in rates of strike-slip faulting: Revisiting the Cherchen River site along the Altyn Tagh Fault, NW China, *Earth Planet. Sci. Lett.*, *254*, 239–255, doi:10.1016/j.epsl.2006.09.015.

Cowgill, E., R. D. Gold, X. H. Chen, X. F. Wang, J. R. Arrowsmith, and J. Southon (2009), Low Quaternary slip rate reconciles geodetic and geologic rates along the Altyn Tagh Fault, northwestern Tibet, *Geology*, *37*, 647–650, doi:10.1130/G25623A.1.

Darby, B. J., B. D. Ritts, Y. J. Yue, and Q. R. Meng (2005), Did the Altyn Tagh fault extend beyond the Tibetan Plateau?, *Earth Planet. Sci. Lett.*, *240*, 425–435, doi:10.1016/j.epsl.2005.09.011.

Dayem, K. E., P. Molnar, M. K. Clark, and G. A. Houseman (2009), Far-field lithospheric deformation in Tibet during continental collision, *Tectonics*, *28*, TC6005, doi:10.1029/2008TC002344.

Deenen, M. H. L., C. G. Langereis, D. J. J. van Hinsbergen, and A. J. Biggin (2011), Geomagnetic secular variation and the statistics of palaeomagnetic directions, *Geophys. J. Int.*, *186*, 509–520, doi:10.1111/j.1365-246X.2011.05050.x.

Deng, C. L., R. X. Zhu, K. L. Verosub, M. J. Singer, and N. J. Vidic (2004), Mineral magnetic properties of loess/paleosol couplets of the central loess plateau of China over the last 1.2 Myr, *J. Geophys. Res.*, *109*, B01103, doi:10.1029/2003JB002532.

Dupont-Nivet, G., R. F. Butler, A. Yin, and X. Chen (2003), Paleomagnetism indicates no Neogene vertical axis rotations of the northeastern Tibetan Plateau, *J. Geophys. Res.*, *108*(B8), 2386, doi:10.1029/2003JB002399.

Dupont-Nivet, G., R. F. Butler, A. Yin, and X. Chen (2002), Paleomagnetism indicates no Neogene rotation of the Qaidam Basin in North Tibet during Indo-Asian Collision, *Geology*, *30*(3), 263–266, doi:10.1130/0091-7613(2002)030<0263:PINNRO>2.0.CO;2.

Dupont-Nivet, G., B. K. Horton, R. F. Butler, J. Wang, J. Zhou, and G. L. Waanders (2004a), Paleogene clockwise tectonic rotation of the Xining-Lanzhou region, northeastern Tibetan Plateau, *J. Geophys. Res.*, *109*, B04401, doi:10.1029/2003JB002620.

Dupont-Nivet, G., D. Robinson, R. F. Butler, A. Yin, and H. J. Melosh (2004b), Concentration of crustal displacement along a weak Altyn Tagh fault: Evidence from paleomagnetism of the northern Tibetan Plateau, *Tectonics*, *23*, TC1020, doi:10.1029/2002TC001397.

Dupont-Nivet, G., S. Dai, X. M. Fang, W. Krijgsman, V. Erens, M. Reitsma, and C. Langereis (2008), Timing and distribution of tectonic rotations in the northeastern Tibetan Plateau, in *Investigations into the Tectonics of the Tibetan Plateau, Geol. Soc. Am. Bull. Spec. 444*, edited by B. C. Burchfiel and E. Wang, pp. 73–97, Geol. Soc. of Am., Boulder, Colo., doi:10.1130/2008.2444(05).

Duvall, A. R., and M. K. Clark (2010), Dissipation of fast strike-slip faulting within and beyond northeastern Tibet, *Geology*, *38*(3), 223–226, doi:10.1130/G30711.1.

Elliott, J. R., J. Biggs, B. Parsons, and T. J. Wright (2008), InSAR slip rate determination on the Altyn Tagh Fault, northern Tibet, in the presence of topographically correlated atmospheric delays, *Geophys. Res. Lett.*, *35*, L12309, doi:10.1029/2008GL033659.

England, P., and G. Houseman (1986), Finite strain calculations of continental deformation: 2. Comparison with the India–Asia collision, *J. Geophys. Res.*, *91*, 3664–3676, doi:10.1029/JB091iB03p03664.

Enkin, R. J., Z. Y. Yang, Y. Chen, and V. Courtillot (1992), Paleomagnetic constraints on the geodynamic history of the major blocks of China from the Permian to the present, *J. Geophys. Res.*, *97*, 13,953–13,989, doi:10.1029/92JB00648.

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Fang, X. M., C. Garzione, R. Van der Voo, J. Li, and M. Fan (2003), Flexural subsidence by 29 Ma on the NE edge of Tibet from the magnetostratigraphy of Linxia basin, China, *Earth Planet. Sci. Lett.*, *210*, 545–560, doi:10.1016/S0012-821X(03)00142-0.

Fang, X. M., W. L. Zhang, Q. Q. Meng, J. P. Gao, X. M. Wang, J. King, C. H. Song, S. Dai, and Y. F. Miao (2007), High-resolution magnetostratigraphy of the Neogene Huaitoutala section in the eastern Qaidam Basin on the NE Tibetan Plateau, Qinghai Province, China and its implication on tectonic uplift of the NE Tibetan Plateau, *Earth Planet. Sci. Lett.*, *258*, 293–306, doi:10.1016/j.epsl.2007.03.042.

Fisher, R. (1953), Dispersion on a sphere, *Proc. R. Soc. London, Ser. A*, *217*, 295–305, doi:10.1098/rspa.1953.0064.

Flynn, L. J., W. Downs, N. Opdyke, K. Huang, E. Lindsay, J. Ye, G. Xie, and X. Wang (1999), Recent advances in the small mammal biostratigraphy and magnetostratigraphy of Lanzhou basin, *Chin. Sci. Bull.*, *44*, 105–118.

Fu, B., and Y. Awata (2007), Displacement and timing of left-lateral faulting in the Kunlun fault zone, northern Tibet, inferred from geologic and geomorphic features, *J. Asian Earth Sci.*, *29*, 253–265, doi:10.1016/j.jseaes.2006.03.004.

Fu, B., Y. Awata, J. Du, and W. He (2005), Late Quaternary systematic stream offsets caused by repeated large seismic events along the Kunlun Fault, northern Tibet, *Geomorphology*, *71*, 278–292, doi:10.1016/j.geomorph.2005.03.001.

Gan, W. J., P. Z. Zhang, Z. K. Shen, Z. J. Niu, M. Wang, Y. G. Wan, D. M. Zhou, and J. Cheng (2007), Present-day crustal motion within the Tibetan Plateau inferred from GPS measurements, *J. Geophys. Res.*, *112*, B08416, doi:10.1029/2005JB004120.

Halim, N., J. P. Cogné, Y. Chen, R. Atasiei, J. Besse, V. Courtillot, S. Gilder, J. Marcoux, and R. L. Zhao (1998), New Cretaceous and Early Tertiary paleomagnetic results from Xining-Lanzhou basin, Kunlun and Qiangtang blocks, China: Implications on the geodynamic evolution of Asia, *J. Geophys. Res.*, *103*(B9), 21,025–21,045.

Heermance, R. V., A. Pullen, P. Kapp, C. N. Garzione, S. Bogue, L. Ding, and P. Song (2013), Climatic and tectonic controls on sedimentation and erosion during the Pliocene-Quaternary in the Qaidam Basin (China), *Bull. Geol. Soc. Am.*, *125*(5–6), 833–856, doi:10.1130/b30748.1.

Huo, G. M. (Ed.) (1990), *Petroleum Geology of China: Oilfields in Qinghai and Xizang [in Chinese]*, Chin. Pet. Ind. Press, Beijing.

Jordan, T. A., and A. B. Watts (2005), Gravity anomalies, flexure and the elastic thickness structure of the India Eurasia collisional system, *Earth Planet. Sci. Lett.*, *236*, 732–750, doi:10.1016/j.epsl.2005.05.036.

Kapp, P., J. D. Pelletier, A. Rohrmann, R. Heermance, J. Russell, and L. Ding (2011), Wind erosion in the Qaidam basin, central Asia: Implications for tectonics, paleoclimate, and the source of the Loess Plateau, *GSA Today*, *21*(4/5), 4–10, doi:10.1130/GSATG99A.1.

Kent-Corson, M. L., B. D. Ritts, G. S. Zhuang, P. M. Bovet, S. A. Graham, and C. P. Chamberlain (2009), Stable isotopic constraints on the tectonic, topographic, and climatic evolution of the northern margin of the Tibetan Plateau, *Earth Planet. Sci. Lett.*, *282*, 158–166, doi:10.1016/j.epsl.2009.03.011.

King, R. F. (1955), The remanent magnetism of artificially deposited sediments, *Geophys. J.*, *7*, 115–134, doi:10.1111/j.1365-246X.1955.tb06558.x.

Kirschvink, J. L. (1980), The least-squares line and plane and the analysis of paleomagnetic data, *Geophys. J.*, *62*, 699–718, doi:10.1111/j.1365-246X.1980.tb02601.x.

Li, H., J. Van der Woerd, P. Tapponnier, Y. Klinger, X. X. Qi, J. S. Yang, and Y. T. Zhu (2005), Slip rate on the Kunlun fault at Hongshui Gou, and recurrence time of great events comparable to the 14/11/2001, Mw∼7.9 Kokoxili earthquake, *Earth Planet. Sci. Lett.*, *237*, 285–299, doi:10.1016/j.epsl.2005.05.041.

Lin, A., and J. Guo (2008), Nonuniform slip rate and millennial recurrence interval of large earthquakes along the eastern segment of the Kunlun fault, northern Tibet, *Bull. Seismol. Soc. Am.*, *98*, 2866–2878, doi:10.1785/0120070193.

Lin, J., and D. R. Watts (1988), Palaeomagnetic results from the Tibetan plateau, *Philos. Trans. R. Soc. London A*, *327*, 239–262, doi:10.1098/rsta.1988.0128.

Liu, Z., Zhao X., C. Wang, S. Liu, and H. Yi (2003), Magnetostratigraphy of Tertiary sediments from the Hoh Xil Basin: Implications for the Cenozoic tectonic history of the Tibetan Plateau, *Geophys. J. Int.*, *154*, 233–252, doi:10.1046/j.1365-246X.2003.01986.x.

Lu, H. J., and S. F. Xiong (2009), Magnetostratigraphy of the Dahonggou section, northern Qaidam Basin and its bearing on Cenozoic tectonic evolution of the Qilian Shan and Altyn Tagh Fault, *Earth Planet. Sci. Lett.*, *288*, 539–550, doi:10.1016/j.epsl.2009.10.016.

McElhinny, M. W. (1964), Statistical significance of the fold test in paleomagnetism, *Geophys. J. R. Astron. Soc.*, *8*, 338–340, doi:10.1111/j.1365-246X.1964.tb06300.x.

McFadden, P. L., and M. W. McElhinny (1988), The combined analysis of remagnetization circles and direct observations in palaeomagnetism, *Earth Planet. Sci. Lett.*, *87*, 161–172, doi:10.1016/0012-821X(88)90072-6.

Métivier, F., Y. Gaudemer, P. Tapponnier, and B. Meyer (1998), Northeastward growth of the Tibet plateau deduced from balanced reconstruction of two depositional areas: The Qaidam and Hexi Corridor basins, China, *Tectonics*, *17*(6), 823–842, doi:10.1029/98TC02764.

Meyer, B., P. Tapponnier, L. Bourjot, F. Métivier, Y. Gaudemer, G. Peltzer, S. Guo, and Z. Chen (1998), Crustal thickening in the Gansu-Qinghai, lithospheric mantle subduction, oblique and strike-slip controlled growth of the Tibet Plateau, *Geophys. J. Int.*, *135*, 1–47, doi:10.1046/j.1365-246X.1998.00567.x.

Pan, Y. X., R. X. Zhu, S. K. Banerjee, J. Gill, and Q. Williams (2000), Rock-magnetic properties related to thermal-treatment of siderite: Behavior and interpretation, *J. Geophys. Res.*, *2000*, 105, 783–794, doi:10.1029/1999JB900358.

Rieser, A. B., Y. J. Liu, J. Genser, F. Neubauer, R. Handler, G. Friedl, and X. H. Ge (2006a), $^{40}Ar/^{39}Ar$ ages of detrital white mica constrain the Cenozoic development of the intracontinental Qaidam Basin, China, *Geol. Soc. Am. Bull.*, *118*, 1522–1534, doi:10.1130/B25962.1.

Rieser, A. B., Y. J. Liu, J. Genser, F. Neubauer, R. Handler, and X. H. Ge (2006b), Uniform permian $^{40}Ar/^{39}Ar$ detrital mica ages in the eastern Qaidam Basin (NW China): Where is the source?, *Terra Nova*, *18*, 79–87, doi:10.1111/j.1365-3121.2005.00666.x.

Roberts, A. P., Y. Cui, and K. L. Verosub (1995), Wasp-waisted hysteresis loops: Mineral magnetic characteristics and discrimination of component in mixed magnetic systems, *J. Geophys. Res.*, *100*, 17,909–17,924, doi:10.1029/95JB00672.

Robinson, D. M., G. Dupont-Nivet, G. E. Gehrels, and Y. Zhang (2003), The Tula uplift, northwestern China: Evidence for regional tectonism of the northern Tibetan Plateau during late Mesozoic-early Cenozoic time, *Bull. Geol. Soc. Am.*, *115*(1), 35–47, doi:10.1130/0016-7606(2003)1152.0.CO;2.

Rumelhart, P. E., A. Yin, E. Cowgill, R. Butler, Q. Zhang, and X. F. Wang (1999), Cenozoic vertical-axis rotation of the Altyn Tagh fault system, *Geology*, *27*(9), 819–822, doi:10.1130/0091-7613(1999)027<0819:CVAROT>2.3.CO;2.

Shen, Z. K., M. Wang, Y. Li, D. D. Jackson, A. Yin, D. Dong, and P. Fang (2001), GPS study of crustal deformation associated with the Altyn Tagh fault system, *J. Geophys. Res.*, *106*, 30,607–30,621, doi:10.1029/2001JB000349.

Si, J., and R. Van der Voo (2001), Too-low magnetic inclinations in central Asia: An indication of a long-term Tertiary non-dipole field?, *Terra Nova*, *13*, 471–478, doi:10.1046/j.1365-3121.2001.00383.x.

Song, B., K. Zhang, J. Lu, C. Wang, and Y. Xu (2013), The middle Eocene to early Miocene integrated sedimentary record in the Qaidam Basin and its implications for paleoclimate and early Tibetan Plateau uplift, *Can. J. Earth Sci.*, *50*(2), 183–196, doi:10.1139/cjes-2012-0048.

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Sun, Z. C., X. J. Feng, D. M. Li, F. Yang, Y. H. Qu, and H. J. Wang (1999), Cenozoic ostracoda and palaeoenvironments of the northeastern Tarim Basin, western China, *Palaeogeogr. Palaeoclimatol. Palaeoecol.*, *148*(1–3), 37–50, doi:10.1016/S0031-0182(98)00174-6.

Sun, Z. M., Z. Y. Yang, J. L. Pei, X. H. Ge, X. S. Wang, T. S. Yang, W. M. Li, and S. H. Yuan (2005), Magnetostratigraphy of Paleogene sediments from northern Qaidam Basin, China: Implications for tectonic uplift and block rotation in northern Tibetan Plateau, *Earth Planet. Sci. Lett.*, *237*, 635–646, doi:10.1016/j.epsl.2005.07.007.

Sun, Z. M., Z. Y. Yang, J. L. Pei, T. S. Yang, and X. S. Wang (2006), New Early Cretaceous paleomagnetic data from volcanic and red beds of the eastern Qaidam Block and its implications for tectonics of Central Asia, *Earth Planet. Sci. Lett.*, *243*, 268–281, doi:10.1016/j.epsl.2005.12.016.

Tauxe, L. (2010), Essentials of Paleomagnetism, pp. 1–489, University of California Press, Berkeley, Calif.

Tauxe, L., and D. V. Kent (2004), A simplified statistical model for the geomagnetic field and the detection of shallow bias in paleomagnetic inclinations: Was the ancient magnetic field dipolar, *Geophys. Monogr. Ser.*, *145*, 101–115, doi:10.1029/145GM08.

Van der Woerd, J., F. J. Ryerson, P. Tapponnier, Y. Gaudemer, R. Finkel, A. S. Meriaux, M. Caffee, G. Zhao, and Q. He (1998), Holocene left-slip rate determined by cosmogenic surface dating on the Xidatan segment of the Kunlun Fault (Qinghai, China), *Geology*, *26*, 695–698, doi:10.1130/0091-7613(1998)026<0695:HLSRDB>2.3.CO;2.

Van Hinsbergen, D. J. J., P. Kapp, G. Dupont-Nivet, P. C. Lippert, P. G. DeCelles, and T. H. Torsvik (2011), Restoration of Cenozoic deformation in Asia and the size of Greater India, *Tectonics*, *30*, TC5003, doi:10.1029/2011TC002908.

Van Velzen, A. J., and M. J. Dekkers (1999), The incorporation of thermal methods in mineral magnetism of loess-paleosol sequences: A brief overview, *Chin. Sci. Bull.*, *44*(S1), 53–63.

Wallace, K., G. Yin, and R. Bilham (2004), Inescapable slow slip on the Altyn Tagh Fault, *Geophys. Res. Lett.*, *31*, L09613, doi:10.1029/2004GL019724.

Wang, F., C. H. Lo, Q. Li, M. W. Yeh, J. L. Wan, D. Zheng, and E. Q. Wang (2004), Onset timing of significant unroofing around Qaidam basin, northern Tibet, China: Constraints from $^{40}Ar/^{39}Ar$ and FT thermochronology on granitoids, *J. Asian Earth Sci.*, *24*, 59–69, doi:10.1016/j.jseaes.2003.07.004.

Wang, Q., et al. (2001), Present-day crustal deformation in China constrained by Global Positioning System measurements, *Science*, *294*, 574–577, doi:10.1126/science.1063647.

Wang, Q. M., and M. P. Coward (1993), The Chaidam Basin, NW China: Formation and hydrocarbon potential, *J. Pet. Geol.*, *13*, 93–112, doi:10.1111/j.1747-5457.1990.tb00255.x.

Watson, G. S., and R. J. Enkin (1993), The fold test in paleomagnetism as a parameter estimation problem, *Geophys. Res. Lett.*, *20*, 2135–2137, doi:10.1029/93GL01901.

Westphal, M. (1993), Did a large departure from the geocentric axial dipole hypothesis occur during the Eocene? Evidence from the magnetic polar wander path of Eurasia, *Earth Planet. Sci. Lett.*, *117*, 15–28, doi:10.1016/0012-821X(93)90114-O.

Wu, L., A. C. Xiao, S. F. Yang, L. Q. Wang, L. G. Mao, L. Wang, Y. P. Dong, and B. Xu (2012), Two-stage evolution of the Altyn Tagh Fault during the Cenozoic: New insight from provenance analysis of a geological section in NW Qaidam basin, NW China, *Terra Nova*, *24*, 387–395, doi:10.1111/j.1365-3121.2012.01077.x.

Yan, M. D., R. Van der Voo, X. M. Fang, J. M. Parés, and D. K. Rea (2006), Paleomagnetic evidence for a mid-Miocene clockwise rotation of about 25° of the Guide Basin area in NE Tibet, *Earth Planet. Sci. Lett.*, *241*, 234–247, doi:10.1016/j.epsl.2005.10.013, Proceedings of 9th International Symposium on Ostracada, Shizuoka, Japan.

Yang, F. (1988), Distribution of the brackish-salt water ostracod in northwestern Qinghai Plateau and its geological significance, in *Evolutionary Biology of Ostracoda*, edited by T. Hanai et al., pp. 519–530, doi:10.1016/S0920-5446(08)70205-X, Proceedings of 9th International Symposium on Ostracada, Shizuoka, Japan.

Yang, F., Z. Ma, T. Xu, and S. Ye (1992), A Tertiary paleomagnetic stratigraphic profile in Qaidam basin [in Chinese], *Acta Petrol. Sin.*, *13*, 97–101.

Yin, A., and T. M. Harrison (2000), Geologic evolution of the Himalayan-Tibetan orogen, *Ann. Rev. Earth Planet. Sci.*, *28*, 211–280, doi:10.1146/annurev.earth.28.1.211.

Yin, A., et al. (2002), Tectonic history of the Altyn Tagh fault system in northern Tibet inferred from Cenozoic sedimentation, *Geol. Soc. Am. Bull.*, *114*, 1257–1295, doi:10.1130/0016-7606(2002)114<1257:THOTAT>2.0.CO.

Yin, A., Y. Q. Dang, M. Zhang, M. W. McRivette, W. P. Burgess, and X. H. Chen (2007), Cenozoic tectonic evolution of Qaidam basin and its surrounding regions. (Part 2) Wedge tectonics in southern Qaidam basin and the Eastern Kunlun Range, in *Whence the Mountains? Inquiries into the Evolution of Orogenic Systems: A Volume in Honor of Raymond A. Price, Geol. Soc. Am. Bull. Spec. Pap.*, vol. 433, edited by J. W. Sears, et al., pp. 369–390, The Geological Society of America, Boulder, Colo., doi:10.1130/2007.2433(18).

Yin, A., Y. Dang, L. Wang, W. Jiang, S. Zhou, X. Chen, G. E. Gehrels, and M. W. McRivette (2008a), Cenozoic tectonic evolution of Qaidam basin and its surrounding regions (Part 1): The southern Qilian Shan-Nan Shan thrust belt and northern Qaidam basin, *Geol. Soc. Am. Bull.*, *120*, 813–846, doi:10.1130.B26180.1. The Geological Society of America, Boulder, Colo.

Yin, A., Y. Q. Dang, M. Zhang, X. H. Chen, and M. W. McRivette (2008b), Cenozoic tectonic evolution of the Qaidam basin and its surrounding regions (Part 3): Structural geology, sedimentation, and regional tectonic reconstruction, *Geol. Soc. Am. Bull.*, *120*, 847–876, doi:10.1130/B26232.1.

Yuan, D.-Y., et al. (2013), The growth of northeastern Tibet and its relevance to large-scale continental geodynamics: A review of recent studies, *Tectonics*, *32*, 1358–1370, doi:10.1002/tect.20081.

Yue, Y. J., B. D. Ritts, and S. A. Graham (2001), Initiation and long-term slip history of the Altyn Tagh fault, *Int. Geol. Rev.*, *43*, 1087–1094, doi:10.1080/00206810109465062.

Zhang, P. Z., et al. (2004), Continuous deformation of the Tibetan Plateau from global positioning system data, *Geology*, *32*, 809–812, doi:10.1130/G20554.1.

Zhang, P. Z., P. Molnar, and X. Xu (2007), Late Quaternary and present-day rates of slip along the Altyn Tagh Fault, northern margin of the Tibetan Plateau, *Tectonics*, *26*, TC5010, doi:10.1029/2006TC002014.

Zhang, W. L., E. Appel, X. M. Fang, C. H. Song, and O. Cirpka (2012a), Magnetostratigraphy of deep drilling core SG-1 in the western Qaidam Basin (NE Tibetan Plateau) and its tectonic implications, *Quat. Res.*, *78*, 139–148, doi:10.1016/j.yqres.2012.03.011.

Zhang, W. L., E. Appel, X. M. Fang, M. D. Yan, C. H. Song, and L. W. Cao (2012b), Paleoclimatic implications of magnetic susceptibility in Late Pliocene–Quaternary sediments from deep drilling core SG-1in the western Qaidam Basin (NE Tibetan Plateau), *J. Geophys. Res.*, *117*, B06101, doi:10.1029/2011JB008949.

Zheng, W. J., P. J. Zhang, W. G. He, D. Y. Yuan, Y. X. Shao, D. W. Zheng, W. P. Ge, and W. Min (2013), Transformation of displacement between strike-slip and crustal shortening in the northern margin of the Tibetan Plateau: Evidence from

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decadal GPS measurements and late Quaternary slip rates on faults, *Tectonophysics*, *584*, 267–280, doi:10.1016/j.tecto.2012.01.006.

Zhou, J. X., F. Y. Xu, T. C. Wang, A. F. Cao, and C. M. Yin (2006), Cenozoic deformation history of the Qaidam Basin, NW China: Results from cross-section restoration and implications for Qinghai-Tibet Plateau tectonics, *Earth Planet. Sci. Lett.*, *243*, 195–210, doi:10.1016/j.epsl.2005.11.033.

Zhu, L., T. J. Owens, and G. E. Randall (1995), Lateral variation in crustal structure of the northern Tibetan Plateau inferred from teleseismic receiver functions, *Bull. Seismol. Soc. Am.*, *85*, 1531–1540, doi:10.1016/j.yqres.2012.03.011.

Zhu, L. D., C. S. Wang, H. B. Zheng, F. Xiang, H. S. Yi, and D. Z. Liu (2006), Tectonic and sedimentary evolution of basins in the northeast of Qinghai-Tibet Plateau and their implication for the northward growth of the plateau, *Palaeogeogr. Palaeoclimatol. Palaeoecol.*, *241*, 49–60, doi:0.1016/j.palaeo.2006.06.019.

Zhu, L. P., and D. V. Helmberger (1998), Moho offset across the northern margin of the Tibetan Plateau, *Science*, *281*, 1170–1172, doi:10.1126/science.281.5380.1170.

Zhuang, G. S., J. K. Hourigan, B. D. Ritts, and M. L. Kent-Corson (2011), Cenozoic multiple-phase tectonic evolution of the northern Tibetan plateau: Constraints from sedimentary records from Qaidam basin, Hexi corridor, and Subei basin, northwest China, *Am. J. Sci.*, *311*, 116–152, doi:10.2475/02.2011.02.

Zijderveld, J. D. A. (1967), A.C. demagnetization of rocks: Analysis of results, In *Methods in Paleomagnetism*, edited by D. W. Collinson et al., pp. 254–286, Elsevier, N. Y.

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